

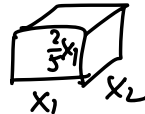
18-52-Q2

Q 98 cm^3  $\$10/\text{cm}^2$ least cost

Solution (a) Formulate a nonlinear program

① length x_1 breadth x_2 let z denote the cost

$$\text{height} = \frac{x_1}{2.5} = \frac{2}{5}x_1$$



$$\text{Min } z = 10 \left[2x_1x_2 + 2 \frac{2}{5}x_1^2 + 2 \frac{2}{5}x_1x_2 \right]$$

$$= 20x_1x_2 + 8x_1^2 + 8x_1x_2$$

$$= 8x_1^2 + 28x_1x_2$$

$$\text{subject to } \begin{cases} x_1x_2 \frac{x_1}{2.5} = 98 \Rightarrow x_1^2x_2 = 245 \\ x_1, x_2 \geq 0 \end{cases}$$

② formulate

$$\text{Min } z = 8x_1^2 + 28x_1x_2$$

$$\text{s.t. } \begin{cases} x_1^2x_2 = 245 \\ x_1, x_2 \geq 0 \end{cases}$$

(b) ① Lagrange multipliers

$$L = 8x_1^2 + 28x_1x_2 + \lambda_1(x_1^2x_2 - 245)$$

$$\begin{cases} \frac{\partial L}{\partial x_1} = 16x_1 + 28x_2 + 2\lambda_1 x_1 x_2 = 0 & (1) \\ \frac{\partial L}{\partial x_2} = 28x_1 + \lambda_1 x_1^2 = 0 & (2) \\ \frac{\partial L}{\partial \lambda} = x_1^2 x_2 - 245 = 0 & (3) \end{cases}$$

$$\text{from (2)} \quad \lambda_1 = \frac{-28x_1}{x_1^2} = \frac{-28}{x_1} \quad (4)$$

$$\begin{aligned} (4) \rightarrow (1) \quad & 16x_1 + 28x_2 + 2x_1 x_2 \left(\frac{-28}{x_1} \right) \\ & = 16x_1 - 28x_2 = 0 \end{aligned}$$

$$\text{So } x_2 = \frac{4}{7} x_1 \quad (5)$$

$$(5) \rightarrow (3) \quad x_1^2 \frac{4}{7} x_1 = 245$$

$$\text{So } x_1 = \sqrt[3]{\frac{245 \times 7}{4}}$$

$$= 7.5405$$

$$x_2 = \frac{4}{7} x_1 = 4.3089$$

$$\lambda_1 = -\frac{28}{x_1} = -3.713$$

$$\textcircled{2} \begin{cases} x_1 = 7.5405 \\ x_2 = 4.3089 \\ \lambda_1 = -3.713 \end{cases}$$

dimensions of the container

length: 7.5405 (cm)

breadth: 4.3089 (cm)

height: 3.0162 (cm)

$$\text{cost} = 8x_1^2 + 28x_1x_2 = 1364.6284$$

\textcircled{3} optimum material cost?

$$h(x) = x_1^2 x_2 - 245$$

$$\frac{\partial h(x)}{\partial x_1} = 2x_1 x_2 \quad \frac{\partial h(x)}{\partial x_2} = x_1^2$$

$$\nabla h(x)^T = \left[\frac{\partial h(x)}{\partial x_1} \quad \frac{\partial h(x)}{\partial x_2} \right]^T = [2x_1 x_2 \quad x_1^2]^T$$

constrain $\nabla h(x)^T \gamma$

$$= [2x_1 x_2 \quad x_1^2] \begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix}$$

$$= 2x_1 x_2 \gamma_1 + x_1^2 \gamma_2 = 0$$

$$\gamma_1 = \frac{-x_1^2}{2x_1 x_2} \gamma_2 = \frac{-x_1}{2x_2} \gamma_2 = -0.8750 \gamma_2 \quad (6)$$

④ Hessian matrix

$$\nabla_{xx}^2 L = \begin{bmatrix} \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} \\ \frac{\partial^2 L}{\partial x_2 \partial x_1} & \frac{\partial^2 L}{\partial x_2^2} \end{bmatrix} \quad \begin{aligned} \frac{\partial L}{\partial x_1} &= 16x_1 + 28x_2 + 2\lambda_1 x_1 x_2 = 0 \\ \frac{\partial L}{\partial x_2} &= 28x_1 + \lambda_1 x_1^2 = 0 \end{aligned}$$

$$= \begin{bmatrix} 16 + 2\lambda_1 x_2 & 28 + 2\lambda_1 x_1 \\ 28 + 2\lambda_1 x_1 & 0 \end{bmatrix} = \begin{bmatrix} -15.9979 & -27.9958 \\ -27.9958 & 0 \end{bmatrix}$$

$$A_1 = -15.9979 < 0$$

$$A_2 = -27.9958^2 < 0$$

So $\nabla_{xx}^2 L$ is a indefinite

$$Y^T \nabla_{xx}^2 L Y = [y_1, y_2] \begin{bmatrix} 16 + 2\lambda_1 x_2 & 28 + 2\lambda_1 x_1 \\ 28 + 2\lambda_1 x_1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= \left[(16 + 2\lambda_1 x_2)y_1 + (28 + 2\lambda_1 x_1)y_2 \quad (28 + 2\lambda_1 x_1)y_1 \right] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= (16 + 2\lambda_1 x_2)y_1^2 + (28 + 2\lambda_1 x_1)y_1 y_2 + (28 + 2\lambda_1 x_1)y_1 y_2$$

$$= (16 + 2\lambda_1 x_2)y_1^2 + 2(28 + 2\lambda_1 x_1)y_1 y_2$$

$$= (16 + 2 \times (-3.713) \times 4.3089) y_1^2 + 2(28 + 2 \times (-3.713) \times 7.5405) y_1 y_2$$

$$= -15.9979 y_1^2 - 55.9915 y_1 y_2$$

$$\text{from (6)} \quad y_1 = -0.8750 y_2$$

$$\begin{cases} x_1 = 7.5405 \\ x_2 = 4.3089 \\ \lambda_1 = -3.713 \end{cases}$$

$$y_2 = -1.1429 y_1$$

$$Y^T \nabla_{xx}^2 L Y = -15.9979 y_1^2 - 55.9915 y_1 y_2$$

$$= 47.9924 y_1^2$$

$$\text{So for } \forall y \neq 0, \quad Y^T \nabla_{xx}^2 L Y > 0$$

So the result is optimal, the Z is the minimum cost